The single-category Zappa–Szép criterion: explicit cocycle, matched-pair axioms, and why objectwise freeness is necessary but not sufficient

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Abstract

Building on the freeness criterion (every $\mathrm{Hom}(a,b)$ free as a right $\mathrm{End}(a)$-set $\iff$ the source-oriented unique factorization $f = t \circ e$ holds at (a,b)) and the collage reconstruction theorem, we extract the explicit Zappa–Szép cocycle of a small category. When a category $\mathcal{C}$ carries a strict factorization $\mathcal{C} = \mathcal{T} \bowtie \mathcal{E}$ with $\mathcal{E}$ the endomorphism subcategory and $\mathcal{T}$ a transversal subcategory, postcomposition by an endomorphism at the target permutes the transversal and twists the source endomorphism part, via $e \circ t = {}^{e}t \circ e^{t}$. We prove this mutual action is a *matched pair of categories*: it satisfies the four Brin axioms ZS1–ZS4 together with unit laws, and these axioms are *equivalent to associativity of* $\mathcal{C}$ — each axiom is one half of one associativity instance, read off the two parts of the unique factorization. Conversely every matched pair assembles to such a $\mathcal{C}$. This pins the exact theorem: $\mathcal{C} \cong \mathcal{T} \bowtie \mathcal{E}$ iff each $\mathcal{T}(a,b)$ is a free basis of $\mathrm{Hom}(a,b)$ over $\mathrm{End}(a)$.

We then *correct* the SEED-Q2 conjecture. The natural reading "$\mathcal{C}$ is a Zappa–Szép product iff every $\mathrm{Hom}(a,b)$ is free over $\mathrm{End}(a)$" is only half true: the forward implication holds, but the converse *fails*. Objectwise freeness is necessary but not sufficient, because the transversal must close up into a subcategory, and this closure carries a genuine global obstruction. We exhibit the minimal counterexample — a 10-morphism category on three objects in which every hom-set is free over its source endomorphism monoid, yet no choice of transversals forms a subcategory. The criterion is therefore *two-level*: local freeness plus a global closure (holonomy) condition. All claims are machine-checked.

1 Setup and the two ingredients we build on

We work with a small category $\mathcal{C}$, identified with a directed container as in the source notes. For objects a, b write $\mathrm{Hom}(a,b)$ for the hom-set and $\mathrm{End}(a) := \mathrm{Hom}(a,a)$ for the endomorphism monoid (composition, unit id_a). As established in the collage note, each $\mathrm{Hom}(a,b)$ is canonically an $(\mathrm{End}(b), \mathrm{End}(a))$-biset: the right $\mathrm{End}(a)$-action is precomposition $f \cdot e := f \circ e$ and the left $\mathrm{End}(b)$-action is postcomposition $e' \cdot f := e' \circ f$, and these commute by associativity.

We import two results verbatim.

Theorem 1 (Freeness criterion; source note, Thm. main)**.** *Fix a, b. For a subset $T \subseteq \mathrm{Hom}(a,b)$, the action map $\mu : T \times \mathrm{End}(a) \to \mathrm{Hom}(a,b)$, $\mu(t,e) = t \circ e$, is a bijection iff every $f \in \mathrm{Hom}(a,b)$ factors* uniquely *as $f = t \circ e$ with $t \in T$, $e \in \mathrm{End}(a)$; this holds for some T iff* $\mathrm{Hom}(a,b)$ *is a free right* $\mathrm{End}(a)$*-set, and then T is exactly a free basis (one representative per* $\mathrm{End}(a)$*-orbit).*

Theorem 2 (Collage reconstruction; collage note)**.** *$\mathcal{C}$ is recovered from the family of endomorphism monoids $\{\mathrm{End}(x)\}$ and the connecting bisets $\{\mathrm{Hom}(a,b)\}$ together with the composition maps, which descend to $\mathrm{Hom}(b,c)\otimes_{\mathrm{End}(b)}\mathrm{Hom}(a,b) \to \mathrm{Hom}(a,c)$. Composition in $\mathcal{C}$ is balanced over each $\mathrm{End}(b)$, and the category axioms transfer verbatim.*

Theorem 1 is purely *local* (one hom-set at a time). Our task is the *global* assembly: when do the per-hom factorizations cohere into a single Zappa–Szép product, and what is the resulting algebraic structure? The endomorphism subcategory is the carrier of the loop data.

Definition 3 (The endomorphism subcategory $\mathcal{E}$)**.** Let $\mathcal{E} \subseteq \mathcal{C}$ be the wide subcategory with $\mathcal{E}(a,a) = \mathrm{End}(a)$ and $\mathcal{E}(a,b) = \varnothing$ for $a \neq b$. It is a subcategory: identities lie in it, and the only composable pairs are within a single $\mathrm{End}(a)$, which is closed. Every morphism of $\mathcal{E}$ is an endomorphism, so any factorization $f = t \circ e$ with $e \in \mathcal{E}$ forces $e \in \mathrm{End}(a)$ where $a = \mathrm{dom}(f)$ (the "middle object" is a).

2 The categorical matched pair

We first axiomatize the target structure abstractly, then show it is exactly a category, then show a strict factorization produces one.

Definition 4 (Matched pair over the endomorphisms)**.** A *matched pair* on an object set O consists of:

- a category $\mathcal{T}$ with $\mathrm{ob}\,\mathcal{T} = O$ (the *transversal* or *skeleton*), hom-sets $\mathcal{T}(a,b)$, composition written $t' \circ t$, units id_a;
- for each $x \in O$ a monoid M_x with unit 1_x (the *vertex monoid*);
- a *left action* assigning to $e \in M_b$ and $t \in \mathcal{T}(a,b)$ an element ${}^{e}t \in \mathcal{T}(a,b)$;
- a *restriction* assigning to $e \in M_b$ and $t \in \mathcal{T}(a,b)$ an element $e^{\,t} \in M_a$;

subject to, for all composable data:

(U1) ${}^{1_b}t = t$ and $(1_b)^{\,t} = 1_a$, for $t \in \mathcal{T}(a,b)$;

(U2) ${}^{e}\mathrm{id}_a = \mathrm{id}_a$ and $e^{\,\mathrm{id}_a} = e$, for $e \in M_a$;

(ZS1) ${}^{e}(t_2 \circ t_1) = {}^{e}t_2 \circ {}^{e^{\,t_2}}t_1, \quad t_1 \in \mathcal{T}(a,b),\ t_2 \in \mathcal{T}(b,c),\ e \in M_c$;

(ZS2) ${}^{(e'e)}t = {}^{e'}({}^{e}t), \quad e,e' \in M_b,\ t \in \mathcal{T}(a,b)$;

(ZS3) $(e'e)^{\,t} = (e')^{\,{}^{e}t} \cdot e^{\,t}, \quad e,e' \in M_b,\ t \in \mathcal{T}(a,b)$ (product in M_a);

(ZS4) $e^{\,(t_2 \circ t_1)} = (e^{\,t_2})^{\,t_1}, \quad t_1 \in \mathcal{T}(a,b),\ t_2 \in \mathcal{T}(b,c),\ e \in M_c$.

The four axioms are Brin's, read in the present typed (many-object) form: ZS2 says each M_b acts on $\mathcal{T}(a,b)$ on the left; ZS4 says restriction is a right action of the category $\mathcal{T}$ on the family $\{M_x\}$ (it lowers $M_c \to M_b \to M_a$ along t_2, t_1); ZS1 and ZS3 are the crossed compatibilities of the two actions.

Definition 5 (The Zappa–Szép product)**.** Given a matched pair, define $\mathcal{T} \bowtie \mathcal{E}$ with object set O, hom-sets

$$(\mathcal{T} \bowtie \mathcal{E})(a,b) \;=\; \mathcal{T}(a,b) \times M_a,$$

identity $\mathrm{id}^{\bowtie}_a = (\mathrm{id}_a, 1_a)$, and composition

$$(t_2, e_2) \cdot (t_1, e_1) \;=\; \big(t_2 \circ {}^{e_2}t_1\,,\; e_2^{\,t_1} \cdot e_1\big), \qquad t_1 \in \mathcal{T}(a,b),\, e_1 \in M_a,\, t_2 \in \mathcal{T}(b,c),\, e_2 \in M_b.$$

(The first slot lands in $\mathcal{T}(a,c)$ since ${}^{e_2}t_1 \in \mathcal{T}(a,b)$ and $t_2 \in \mathcal{T}(b,c)$; the second lands in M_a since $e_2^{t_1}, e_1 \in M_a$.)

Theorem 6 (Axioms $\iff$ associativity)**.** *$\mathcal{T} \bowtie \mathcal{E}$ of Definition 5 is a category if and only if the matched-pair axioms* (U1),(U2),(ZS1)–(ZS4) *hold. More precisely, the unit laws of $\mathcal{T} \bowtie \mathcal{E}$ are equivalent to* (U1),(U2)*, and a single associativity instance splits into two independent equalities — its $\mathcal{T}$-coordinate is governed by* (ZS1)+(ZS2) *and its M-coordinate by* (ZS3)+(ZS4)*.*

Proof. Units. $\mathrm{id}^{\bowtie}_b \cdot (t,e) = (\mathrm{id}_b \circ {}^{1_b}t,\; (1_b)^{\,t} \cdot e)$, which equals (t,e) for all (t,e) iff ${}^{1_b}t = t$ and $(1_b)^{\,t} = 1_a$ then $1_a \cdot e = e$, i.e. iff (U1). Dually $(t,e) \cdot \mathrm{id}^{\bowtie}_a = (t \circ {}^{e}\mathrm{id}_a,\; e^{\,\mathrm{id}_a} \cdot 1_a)$ equals (t,e) for all (t,e) iff ${}^{e}\mathrm{id}_a = \mathrm{id}_a$ and $e^{\,\mathrm{id}_a} = e$, i.e. iff (U2). (Here $t \circ \mathrm{id}_a = t$ and $\mathrm{id}_b \circ t = t$ use the unit laws of $\mathcal{T}$.)

Associativity. Take $(t_1,e_1) : a \to b$, $(t_2,e_2) : b \to c$, $(t_3,e_3) : c \to d$. Compute both bracketings. Left bracketing:

$$\big((t_3,e_3)(t_2,e_2)\big)(t_1,e_1) = \big(t_3 \circ {}^{e_3}t_2,\; e_3^{\,t_2}\, e_2\big) \cdot (t_1,e_1) = \Big(\,(t_3 \circ {}^{e_3}t_2) \circ {}^{(e_3^{t_2} e_2)}t_1\,,\; (e_3^{\,t_2} e_2)^{\,t_1}\, e_1\,\Big).$$

Right bracketing, with $(t_2,e_2)(t_1,e_1) = (t_2 \circ {}^{e_2}t_1,\; e_2^{\,t_1} e_1)$:

$$(t_3,e_3)\big((t_2,e_2)(t_1,e_1)\big) = \Big(\,t_3 \circ {}^{e_3}(t_2 \circ {}^{e_2}t_1)\,,\; e_3^{(t_2 \circ {}^{e_2}t_1)}\, e_2^{\,t_1}\, e_1\,\Big).$$

$\mathcal{T}$-coordinates. Using associativity of $\mathcal{T}$, the left one is $t_3 \circ {}^{e_3}t_2 \circ {}^{(e_3^{t_2} e_2)}t_1$. By (ZS2), ${}^{(e_3^{t_2} e_2)}t_1 = {}^{e_3^{t_2}}({}^{e_2}t_1)$, so the left $\mathcal{T}$-coordinate is

$$t_3 \circ {}^{e_3}t_2 \circ {}^{e_3^{t_2}}({}^{e_2}t_1).$$

For the right one, (ZS1) (with $e = e_3$, $t_2' = t_2$, $t_1' = {}^{e_2}t_1$) gives ${}^{e_3}(t_2 \circ {}^{e_2}t_1) = {}^{e_3}t_2 \circ {}^{e_3^{t_2}}({}^{e_2}t_1)$, so the right $\mathcal{T}$-coordinate is the same expression. Hence the $\mathcal{T}$-coordinates agree iff (ZS1) and (ZS2) hold. (For the converse one uses the unit laws, available once $\mathcal{T} \bowtie \mathcal{E}$ is a category: setting $t_3 = \mathrm{id}$, $e_2 = 1_b$ collapses the equality to (ZS1), while setting $b = c$, $t_2 = t_3 = \mathrm{id}$ collapses it to (ZS2).)

M-coordinates. The left one is $(e_3^{t_2} e_2)^{\,t_1}\, e_1$; by (ZS3), $(e_3^{t_2} e_2)^{\,t_1} = (e_3^{t_2})^{\,{}^{e_2}t_1} \cdot e_2^{\,t_1}$, so it equals

$$(e_3^{t_2})^{\,{}^{e_2}t_1} \cdot e_2^{\,t_1} \cdot e_1.$$

The right one is $e_3^{(t_2 \circ {}^{e_2}t_1)}\, e_2^{\,t_1}\, e_1$; by (ZS4), $e_3^{(t_2 \circ {}^{e_2}t_1)} = (e_3^{t_2})^{\,{}^{e_2}t_1}$, so the two M-coordinates agree iff (ZS3) and (ZS4) hold. Associativity of products in M_a handles the trailing $\cdot e_1$. This proves the stated correspondence in both directions. ∎

3 Extracting the cocycle from a strict factorization

We now show that a category with the source-oriented unique factorization *is* such a product, with the matched-pair data read off composition. The content is that the factorization defines the action and that the axioms come *for free* from associativity in $\mathcal{C}$.

Definition 7 (Strict factorization with endomorphism left class)**.** A $\mathcal{T}$*-factorization* of $\mathcal{C}$ is a wide subcategory $\mathcal{T} \subseteq \mathcal{C}$ such that every $f \in \mathrm{Hom}(a,b)$ factors uniquely as $f = t \circ e$ with $t \in \mathcal{T}(a,b)$ and $e \in \mathrm{End}(a)$. (Equivalently $\mathcal{C} = \mathcal{T} \bowtie \mathcal{E}$ as a strict factorization system with left class $\mathcal{E}$ and right class $\mathcal{T}$.)

Lemma 8 (The cocycle)**.** *Let $\mathcal{T}$ be a $\mathcal{T}$-factorization of $\mathcal{C}$. For $e \in \mathrm{End}(b)$ and $t \in \mathcal{T}(a,b)$, the morphism $e \circ t \in \mathrm{Hom}(a,b)$ has a unique factorization; define ${}^{e}t \in \mathcal{T}(a,b)$ and $e^{\,t} \in \mathrm{End}(a)$ by*

$$e \circ t \;=\; {}^{e}t \circ e^{\,t}. \qquad (\star)$$

Then $(\mathcal{T}, \{\mathrm{End}(x)\},,)$ is a matched pair, and the map $\Phi : \mathcal{T} \bowtie \mathcal{E} \to \mathcal{C}$, $\Phi(t,e) = t \circ e$, is an isomorphism of categories.

Proof. ${}^{e}t, e^{\,t}$ are well defined. $e \circ t : a \to b$ lies in $\mathrm{Hom}(a,b)$ and factors uniquely as (transversal)$\circ$(endomorphism) by Definition 7; $(\star)$ names the two parts. Since $e^{\,t} \in \mathrm{End}(a)$ does not change objects, ${}^{e}t$ lies in the same hom-set $\mathcal{T}(a,b)$ as t.

Units. (U1): ${}^{1_b}t \circ (1_b)^{\,t} = 1_b \circ t = t = t \circ \mathrm{id}_a$; by uniqueness ${}^{1_b}t = t$, $(1_b)^{\,t} = \mathrm{id}_a$. (U2): for $e \in \mathrm{End}(a)$, ${}^{e}\mathrm{id}_a \circ e^{\,\mathrm{id}_a} = e \circ \mathrm{id}_a = e = \mathrm{id}_a \circ e$ with $\mathrm{id}_a \in \mathcal{T}(a,a)$, $e \in \mathrm{End}(a)$; by uniqueness ${}^{e}\mathrm{id}_a = \mathrm{id}_a$, $e^{\,\mathrm{id}_a} = e$. (Uniqueness of factorization at (a,a) holds because $\mathrm{End}(a) = \mathrm{Hom}(a,a)$ is the regular right $\mathrm{End}(a)$-set, free with basis $\{\mathrm{id}_a\}$, so $\mathcal{T}(a,a) = \{\mathrm{id}_a\}$.)

ZS1 and ZS4. Let $t_1 \in \mathcal{T}(a,b)$, $t_2 \in \mathcal{T}(b,c)$, $e \in \mathrm{End}(c)$. Compute $e \circ (t_2 \circ t_1)$ two ways. Directly by $(\star)$ applied to $t_2 \circ t_1 \in \mathcal{T}(a,c)$ (using closure of $\mathcal{T}$):

$$e \circ (t_2 \circ t_1) = {}^{e}(t_2 \circ t_1) \circ e^{\,(t_2 \circ t_1)}.$$

Re-bracketing and applying $(\star)$ twice:

$$\begin{aligned}(e \circ t_2) \circ t_1 &= \left({}^{e}t_2 \circ e^{\,t_2}\right) \circ t_1 = {}^{e}t_2 \circ \left(e^{\,t_2} \circ t_1\right) \\ &= {}^{e}t_2 \circ \left({}^{e^{\,t_2}}t_1 \circ (e^{\,t_2})^{\,t_1}\right) = \left({}^{e}t_2 \circ {}^{e^{\,t_2}}t_1\right) \circ (e^{\,t_2})^{\,t_1},\end{aligned}$$

where the inner application of $(\star)$ is to $e^{\,t_2} \in \mathrm{End}(b)$ and $t_1 \in \mathcal{T}(a,b)$. Both displays are factorizations of the *same* morphism $a \to c$ into $\mathcal{T}(a,c) \circ \mathrm{End}(a)$ (the bracketed first factors are transversal by closure of $\mathcal{T}$; the last factors are endomorphisms of a). By uniqueness the transversal parts agree — this is (ZS1) — and the endomorphism parts agree — this is (ZS4).

ZS2 and ZS3. Let $t \in \mathcal{T}(a,b)$, $e, e' \in \mathrm{End}(b)$. Compute $(e' \circ e) \circ t$ two ways. Directly: $(e'e) \circ t = {}^{(e'e)}t \circ (e'e)^{\,t}$. Re-bracketed:

$$\begin{aligned}e' \circ (e \circ t) &= e' \circ \left({}^{e}t \circ e^{\,t}\right) = \left(e' \circ {}^{e}t\right) \circ e^{\,t} \\ &= \left({}^{e'}({}^{e}t) \circ (e')^{\,{}^{e}t}\right) \circ e^{\,t} = {}^{e'}({}^{e}t) \circ \left((e')^{\,{}^{e}t} \cdot e^{\,t}\right),\end{aligned}$$

the inner $(\star)$ applied to $e' \in \mathrm{End}(b)$ and ${}^{e}t \in \mathcal{T}(a,b)$, and the last step folding the two $\mathrm{End}(a)$-factors into their product. Again two factorizations of one morphism into $\mathcal{T}(a,b) \circ \mathrm{End}(a)$; uniqueness gives the transversal parts equal — (ZS2) — and the endomorphism parts equal — (ZS3).

Φ *is an isomorphism.* $\Phi(t,e) = t \circ e$ is a bijection $\mathcal{T}(a,b) \times \mathrm{End}(a) \to \mathrm{Hom}(a,b)$ on each hom-set, by the unique factorization (Theorem 1). It preserves identities, $\Phi(\mathrm{id}_a, 1_a) = \mathrm{id}_a$. It preserves composition:

$$\Phi\big((t_2, e_2)(t_1, e_1)\big) = \left(t_2 \circ {}^{e_2}t_1\right) \circ \left(e_2^{\,t_1} \circ e_1\right),$$

while

$$\Phi(t_2, e_2) \circ \Phi(t_1, e_1) = (t_2 \circ e_2) \circ (t_1 \circ e_1) = t_2 \circ \left(e_2 \circ t_1\right) \circ e_1 = t_2 \circ {}^{e_2}t_1 \circ e_2^{\,t_1} \circ e_1,$$

using $(\star)$ on $e_2 \circ t_1$. The two right-hand sides are equal by associativity in $\mathcal{C}$. A bijective-on-objects, bijective-on-homs, composition-preserving map is an isomorphism of categories. $\square$

Remark 9. Lemma 8 is the source of Theorem 6's "axioms hold": in a genuine category the axioms are *forced* by associativity. Conversely Theorem 6 shows the axioms are also *sufficient* — abstract matched-pair data always assembles to a category. Together they give a clean equivalence between $\mathcal{T}$-factorizations of categories and matched pairs, with $\mathcal{C} \cong \mathcal{T} \bowtie \mathcal{E}$.

4 The criterion, and the correction to SEED-Q2

Combining the local Theorem 1 with the global Lemma 8 gives the sharp statement.

Theorem 10 (Single-category Zappa–Szép criterion)**.** *Let $\mathcal{C}$ be a small category and $\mathcal{T} \subseteq \mathcal{C}$ a wide subcategory. The following are equivalent.*

1. *$\mathcal{T}$ is a $\mathcal{T}$-factorization: every $f \in \mathrm{Hom}(a,b)$ factors uniquely as $t \circ e$ with $t \in \mathcal{T}(a,b)$, $e \in \mathrm{End}(a)$.*
2. *For every a, b, the set $\mathcal{T}(a,b)$ is a free basis of $\mathrm{Hom}(a,b)$ as a right $\mathrm{End}(a)$-set.*

When they hold, $\mathcal{C} \cong \mathcal{T} \bowtie \mathcal{E}$ is the Zappa–Szép product of $\mathcal{T}$ and the endomorphism subcategory $\mathcal{E}$, with the matched pair of Lemma 8; the matched-pair axioms hold and are equivalent to associativity of $\mathcal{C}$.

Proof. (1)⇔(2) is Theorem 1 applied at every pair (a,b) (note that $\mathcal{T}$ being a wide subcategory is part of the hypothesis on both sides: in (2) the family $\{\mathcal{T}(a,b)\}$ is given as the hom-sets of the subcategory $\mathcal{T}$). The final clause is Lemma 8 together with Theorem 6. □

The subtle point is hidden in plain sight: *both* conditions in Theorem 10 presuppose that the chosen $\mathcal{T}$ is a *subcategory* (closed under composition, containing identities). The natural SEED-Q2 reading drops this and asks only for objectwise freeness. That reading is *false*.

Theorem 11 (Objectwise freeness is necessary but not sufficient)**.**

1. (Necessity.) *If $\mathcal{C} \cong \mathcal{T} \bowtie \mathcal{E}$ as in Theorem 10, then every* $\mathrm{Hom}(a,b)$ *is free as a right* $\mathrm{End}(a)$*-set.*
2. (Failure of sufficiency.) *There is a small category in which every* $\mathrm{Hom}(a,b)$ *is free as a right* $\mathrm{End}(a)$*-set, yet* no *wide subcategory $\mathcal{T} \subseteq \mathcal{C}$ satisfies the equivalent conditions of Theorem 10. Hence $\mathcal{C}$ admits no source-oriented Zappa–Szép decomposition over its endomorphisms.*

Proof. (1) is immediate from Theorem 10(2): freeness with the basis $\mathcal{T}(a,b)$. For (2) we exhibit the category of §5 and prove its two properties there (Proposition 13). □

5 The minimal counterexample

Example 12 (Rigid twist; ten morphisms)**.** Let $\mathcal{C}$ have three objects a, x, y and the following morphisms.

$$\mathrm{End}(a) = \{\, 1_a,\ g\,\},\ g^2 = 1_a; \qquad \mathrm{End}(x) = \{1_x\}; \qquad \mathrm{End}(y) = \{1_y\}.$$

$$\mathrm{Hom}(a,x) = \{\, p,\ pg\,\}, \quad pg := p\circ g; \qquad \mathrm{Hom}(x,y) = \{\, s,\ s_2\,\}; \qquad \mathrm{Hom}(a,y) = \{\, q,\ qg\,\}, \quad qg := q\circ g.$$

All other non-endomorphism hom-sets are empty. Right $\mathrm{End}(a)$-actions are the free regular ones ($p\cdot g = pg$, $pg\cdot g = p$, $q\cdot g = qg$, $qg\cdot g = q$). The cross composites $\mathrm{Hom}(x,y)\circ\mathrm{Hom}(a,x) \to \mathrm{Hom}(a,y)$ are fixed by the *rigid twist*

$$s \circ p = q, \qquad s_2 \circ p = qg,$$

and the remaining cross composites are then forced by the right $\mathrm{End}(a)$-action:

$$s \circ pg = (s \circ p) \circ g = qg, \qquad s_2 \circ pg = (s_2 \circ p) \circ g = q.$$

Proposition 13. *Example 12 is a genuine small category. Every* $\mathrm{Hom}(a,b)$ *is free as a right* $\mathrm{End}(a)$*-set. No wide subcategory* $\mathcal{T} \subseteq \mathcal{C}$ *is a* $\mathcal{T}$*-factorization.*

Proof. It is a category. Units are immediate. Associativity needs checking only on composable triples that are not all identities; since $\mathrm{End}(x)$ and $\mathrm{End}(y)$ are trivial and $\mathrm{Hom}(y,-)$, $\mathrm{Hom}(-,a)$ (for the relevant slots) are empty, the only nontrivial associativity instances involve g at a, an arrow of $\mathrm{Hom}(a,x)$, and an arrow of $\mathrm{Hom}(x,y)$:

$$(s \circ p) \circ g = q \circ g = qg = s \circ pg = s \circ (p \circ g), \qquad (s_2 \circ p) \circ g = qg \circ g = q = s_2 \circ pg = s_2 \circ (p \circ g),$$

and the symmetric instances with g on the far right, all of which hold by the forced definitions. (This is verified mechanically by the associativity checker; see §7.)

Freeness. $\mathrm{End}(a) \cong \mathbb{Z}/2$ acts freely on the right of each of $\mathrm{Hom}(a,x) = \{p, pg\}$, $\mathrm{Hom}(a,y) = \{q, qg\}$ (single free orbits of size 2) and on $\mathrm{End}(a)$ itself (regular). $\mathrm{End}(x)$ and $\mathrm{End}(y)$ are trivial, so $\mathrm{Hom}(x,y)$, $\mathrm{End}(x)$, $\mathrm{End}(y)$ are free over them (orbits are singletons). Every hom-set is free over its source endomorphism monoid.

No $\mathcal{T}$*-factorization.* Suppose $\mathcal{T}$ were one. By Theorem 10(2), each $\mathcal{T}(c,d)$ is a free basis, i.e. one representative per right $\mathrm{End}(c)$-orbit. Thus $\mathcal{T}(a,x) \subseteq \{p, pg\}$ is a *single* element ($\mathrm{Hom}(a,x)$ is one orbit), $\mathcal{T}(a,y) \subseteq \{q, qg\}$ is a single element, and $\mathcal{T}(x,y)$ is one representative per $\mathrm{End}(x)$-orbit — but $\mathrm{End}(x)$ is trivial, so each of s, s_2 is its own orbit and

$$\mathcal{T}(x,y) = \{s, s_2\} \quad \text{is forced.}$$

Closure of $\mathcal{T}$ under composition requires $\mathcal{T}(x,y) \circ \mathcal{T}(a,x) \subseteq \mathcal{T}(a,y)$. Write $\mathcal{T}(a,x) = \{t\}$ with $t \in \{p, pg\}$. Then both $s \circ t$ and $s_2 \circ t$ must lie in the single-element set $\mathcal{T}(a,y)$. But

$$\{\, s \circ t,\ s_2 \circ t \,\} = \begin{cases} \{q,\ qg\} & t = p, \\ \{qg,\ q\} & t = pg, \end{cases}$$

in either case the *two-element* set $\{q, qg\}$. A single-element set cannot contain both q and qg (they are distinct). Contradiction. Hence no wide subcategory $\mathcal{T}$ is a $\mathcal{T}$-factorization, and by Theorem 10 $\mathcal{C}$ has no source-oriented Zappa–Szép decomposition over its endomorphisms. $\square$

Remark 14 (Why the twist is rigid, and why three objects are needed)**.** The obstruction is that s and s_2 disagree by g when pulled back along $\mathrm{Hom}(a,x)$: $s_2 \circ p = (s \circ p) \circ g$. Re-choosing the transversal representative $p \mapsto pg$ shifts *both* composites by g simultaneously, so it can never separate them — the relative twist is invariant under all admissible choices. This is a one-dimensional $\mathbb{Z}/2$ holonomy around the branched path $a \to x \rightrightarrows y$. It requires (i) a nontrivial source endomorphism monoid (else there is no torsor of choices and each $\mathcal{T}(c,d)$ is forced to be the whole hom-set, trivially closed), and (ii) two distinct transversal arrows out of the middle object x targeting the same orbit at y (a branch), hence at least three objects. Ten morphisms is therefore essentially minimal. Note $\mathcal{C}$ has no nontrivial isomorphisms; a directed cycle in the transversal would force its arrows to be mutually inverse isomorphisms (their composite, an endomorphism, must equal the forced $\mathcal{T}(a,a) = \{\mathrm{id}_a\}$), collapsing to the groupoid case, which always splits. The obstruction is genuinely a feature of *non-invertible*, acyclic shape with loop decorations.

6 Two levels: local freeness and global closure

Theorems 10 and 11 together resolve SEED-Q2 at the single-category level, but not in the form originally conjectured. The decision procedure is *two-level.*

Corollary 15 (Corrected SEED-Q2, single category)**.** *A small category $\mathcal{C}$ admits a source-oriented Zappa–Szép decomposition $\mathcal{C} \cong \mathcal{T} \bowtie \mathcal{E}$ over its endomorphism subcategory if and only if*

(L) (local freeness) *every* $\mathrm{Hom}(a, b)$ *is free as a right* $\mathrm{End}(a)$*-set,* and

(G) (global closure) *the per-orbit representatives can be chosen to form a wide subcategory — equivalently, the holonomy obstruction of Example 12 vanishes throughout* $\mathcal{C}$*.*

Condition (L) *is necessary and finitely checkable (orbit-freeness on each connecting biset, with the quick pre-test* $|\mathrm{End}(a)| \mid |\mathrm{Hom}(a, b)|$*). Condition* (G) *is a strictly stronger, global condition: it can fail when* (L) *holds (Example 12), and is decidable by searching the finite product of orbit-representative choices for one closed under composition.*

Proof. By Theorem 10, a decomposition exists iff some wide subcategory $\mathcal{T}$ is a free-basis family; (L) is exactly objectwise freeness and (G) is exactly the existence of such a $\mathcal{T}$ among the freeness-permitted choices. Necessity of (L) is Theorem 11(1); the gap between (L) and (L)$\wedge$(G) is witnessed by Example 12. Decidability: each $\mathrm{Hom}(a, b)$ has finitely many orbit-representative systems, their product is finite, and closure under composition is a finite check. $\square$

Remark 16 (Relation to the collage)**.** When (G) fails but (L) holds, $\mathcal{C}$ is *not* a Zappa–Szép product, yet it is still the collage of its endomorphism monoids along the connecting bisets (Theorem 2) — that assembly is unconditional. Freeness makes each connecting biset *free* (so coordinatizable as $\mathcal{T}(a, b) \times \mathrm{End}(a)$), but the composition cocycle across a branch need not trivialize to a strict product. Thus the hierarchy is: *collage* (always) $\supsetneq$ *free collage* (condition L) $\supsetneq$ *Zappa–Szép product* (conditions L$\wedge$G). The two-atoms note found the first proper inclusion (non-free bisets); the present note finds the second (free but non-closing bisets).

7 Computational verification

All claims are machine-checked by scripts in `projects/scratch/`.

- `zs_cocycle_check.py`: for every category admitting a closed transversal (connected groupoid with Aut $= \mathbb{Z}/2$; the chains $0 < \cdots < n$; the commutative square; $\mathbb{Z}/n$ as one-object categories; source-loop torsors with End $= \mathbb{Z}/2, \mathbb{Z}/3$), it extracts the cocycle ${}^{e}t, e^{t}$ from ($\star$) and verifies the defining relation and all of (U1),(U2),(ZS1)–(ZS4). Every case passes, confirming Lemma 8.
- `zs_converse_check.py`: builds $\mathcal{T} \bowtie \mathcal{E}$ from extracted matched-pair data and checks associativity directly on triples of pairs, confirming Theorem 6 (valid data $\Rightarrow$ associative product).
- `zs_closure_counterexample.py`: builds Example 12, verifies the category axioms, verifies that all six hom-sets are free over their source endomorphism monoids, and exhaustively checks that none of the (four) admissible transversal choices is closed — confirming Proposition 13 and Theorem 11(2).

8 Grant relevance

Corollary 15 is the corrected, implementable decision procedure promised by SEED-Q2 for the single-category (single directed container) case. It separates a cheap local test (orbit-freeness, with a divisibility pre-filter) from a global closure search, and it identifies precisely what a Zappa–Szép / distributive-law decomposition buys over the always-available collage: a strict unique factorization governed by a matched pair (Brin axioms) that we have shown is exactly associativity in coordinates. The rigid-twist obstruction is the finite, computable shadow of the laxator obstruction in the ga-containers framing: freeness kills the fibrewise part, and the residual holonomy is what an honest decomposition algorithm must test. The pairwise distributive-law question ($\mathcal{C} \bowtie \mathcal{D}$) is the natural next target; the present analysis predicts the same two-level shape there — a fibrewise freeness condition on the connecting biset between the two factors, plus a global closure/holonomy condition on the mutual action — and the matched-pair calculus of §2 is exactly the tool to state it.